

Fundamental Quiz

JOB CONNECTOR AI & FULLSTACK WEB DEVELOPMENT

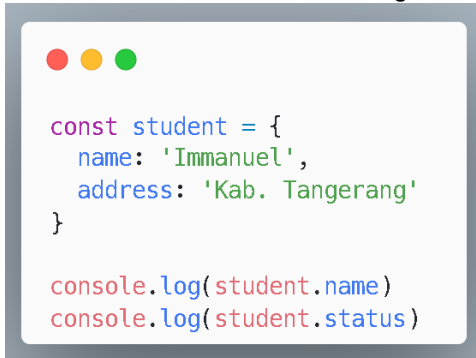
1. **What is the primary purpose of using pseudocode in programming?**
 - a. To write executable code
 - b. To test the code for syntax errors
 - c. To document the code for future reference
 - d. To outline the steps of an algorithm in a simpler language
2. **The difference between `while` and `do while` in Javascript is:**
 - a. In a while loop, the loop will only run if the given condition is false, whereas in a do while loop, it will continue to run continuously even if the given condition is true
 - b. In a while loop, the condition is checked after each iteration before executing the statements inside the while loop, whereas in a do while loop, the condition is checked first before each iteration
 - c. A while loop will only run if the given condition is true, whereas a do while loop will continue to run continuously even if the given condition is false
 - d. In a while loop, the condition is checked before each iteration before executing the statements inside the while loop, whereas in a do while loop, the condition is checked after the iteration has run

3. **What is the result of the following code?**



```
console.log(typeof null);
console.log(typeof 151);
```

- a. Object & Number
 - b. Null & Number
 - c. Undefined & Number
 - d. String & Number
4. **What's the result of the two console.log below?**



```
const student = {
  name: 'Immanuel',
  address: 'Kab. Tangerang'
}

console.log(student.name)
console.log(student.status)
```

- a. Immanuel & Undefined
 - b. Immanuel & Null
 - c. Immanuel & `ReferenceError: properties status is not defined`
 - d. Immanuel & `TypeError: status is not a properties of an Object user`

5. The code, when executed, turns out to be an error. What caused this error?

```
let temp: string[] = "javascript is awesome".split(" ");
for (let i = 0; i < temp.length; i += 1) {
    temp.push(temp[i]);
}
```

- a. Incorrect syntax in the for loop
- b. An infinite loop occurs because of `temp.push(temp[i])`
- c. The condition step in the for loop cannot be `i += 1`
- d. The string in the variable `str` cannot be directly given the `.split(' ')` method

6. Adding a New Property to Each Object Given the following array:

```
const students = [
  { id: 1, name: "John", score: 80 },
  { id: 2, name: "Jane", score: 90 }
];
```

How can we add a new property status with the value "Pass" if the score is ≥ 85 , otherwise "Fail"?

- a. `students.forEach(s => s.status = s.score  $\geq$  85 ? "Pass" : "Fail");`
 - b. `students.map(s => ({ ...s, status: s.score  $\geq$  85 ? "Pass" : "Fail" }));`
 - c. `students.filter(s => s.score  $\geq$  85).map(s => s.status = "Pass");`
 - d. `students.reduce((acc, s) => acc.push({ ...s, status: s.score  $\geq$  85 ? "Pass" : "Fail" }), []);`
7. What is the purpose of the 'super' keyword in class?
- a. To define a new method in the class
 - b. To inherit properties from another class
 - c. To call the parent class's constructor
 - d. To create a new instance of the class
8. Which of the following is a characteristic of a queue data structure?
- a. Last-in, first-out (LIFO)
 - b. Random access
 - c. First-in, first-out (FIFO)
 - d. Sorted order

9. Observe code below:

```
const sentence: string = "Learn Javascript at Purwadhika";
sentence.replace("a", "o");
sentence.split(" ");
sentence.reverse();
console.log(sentence);
```

What is the output from the code above?

- a. ['Purwadhika', 'at', 'Javascript', 'Leorn']
- b. TypeError: sentence.reverse is not a function
- c. Leorn Jovoscript ot Purwodhiko
- d. Leorn Javascript at Purwadhika

10. Consider the following code, which time complexity best describes the code?

- a. $O(n)$
- b. $O(1)$
- c. $O(\log n)$
- d. $O(n^2)$